RUBRICEYE EXAM

Integrated Science & Computing — Mock Examination — Marking Rubric

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Exam Version: RE-01 | Maximum Marks: 35 | Section B: 20 marks (attempt 5 of 7) | Section C: 15 marks (attempt 1 of 2)

SECTION B — Short Answer Questions (20 Marks)

Q2: Attempt Only 5 Questions. Each part below is worth 4 marks.

i. — 4 marks

State Newton's Second Law of Motion and write its mathematical expression.

- Correct statement of the law (net force is proportional to rate of change of momentum, or acceleration is proportional to net force and inversely proportional to mass) — 2 marks
- Correct mathematical expression, $F = ma$, with F, m, and a identified — 2 marks

ii. — 4 marks

Differentiate between RAM and ROM. Give one example of each.

- Correct differentiation: RAM is volatile, read-write working memory; ROM is non-volatile, read-only (or read-mostly) permanent storage — 2 marks
- One valid real-world example of RAM (e.g. DDR4/DDR5 memory module) — 1 mark
- One valid real-world example of ROM (e.g. BIOS/firmware chip) — 1 mark

iii. — 4 marks

Explain the role of the nucleus in a cell.

- Identifies the nucleus as the control centre of the cell — 1 mark
- States that it contains genetic material (DNA/chromosomes) — 1 mark
- Explains that it controls/regulates cell activities, growth, and reproduction — 1 mark
- Overall explanation is clear and coherent, not just a list of disconnected facts — 1 mark

iv. — 4 marks

What is the difference between a series and a parallel circuit? Also draw a diagram.

- Correct textual difference: series circuit has a single current path with the same current throughout; parallel circuit has multiple branches with the same voltage across each — 2 marks
- Correct, clearly labelled diagram showing both a series and a parallel circuit — 2 marks

A diagram-only or largely-diagrammatic answer that correctly conveys both circuit types should receive full credit even without extended prose — judge the underlying concept, not the format.

v. — 4 marks

Name two input devices and two output devices used in a computer system. And what operation do they perform.

- Two valid input devices named (e.g. keyboard, mouse, scanner, microphone) — 1 mark
- Two valid output devices named (e.g. monitor, printer, speaker) — 1 mark
- Correct explanation of what an input device does (accepts data/instructions into the system) — 1 mark
- Correct explanation of what an output device does (presents processed data to the user) — 1 mark

vi. — 4 marks

A force acts on a box on a horizontal surface. Study the given force diagram and identify the forces acting upward, downward, forward and backward. Complete the diagram with proper labeling.

- The question paper's diagram shows a box on a horizontal surface with four arrows: one pointing up, one down, one left, one right. Correct labelling expected:
- Upward arrow correctly identified as the Normal / Reaction force — 1 mark
- Downward arrow correctly identified as Weight / Gravitational force — 1 mark
- Rightward (forward) arrow correctly identified as the Applied / Forward force — 1 mark
- Leftward (backward) arrow correctly identified as Friction / the backward-resisting force — 1 mark

The question explicitly asks the student to complete/label the given diagram, not redraw it from scratch — accept a labelled diagram, a text-only identification of all four forces, or a mix of both, provided all four directions are correctly and unambiguously identified. This is the same diagram/prose fairness principle used throughout grading: do not penalise the answer's format, only its correctness.

vii. — 4 marks

Draw a bar graph representing the following data: Ahmed 45, Sara 20, Ali 35, Khadija 40 (out of a total of 50).

- Correctly labelled axes (students on one axis, marks on the other) — 1 mark
- Bar heights are proportional and consistent with the given data (Ahmed 45, Sara 20, Ali 35, Khadija 40) — 2 marks
- Graph is neat, clearly readable, and each bar is labelled with the correct student name — 1 mark

SECTION C — Structured / Extended Response (15 Marks)

Answer any 1 of Q3 or Q4. Each question is worth 15 marks total (8 + 7).

Q3: Attempt Both Parts of the following Question [8+7]

Part (a) — 8 marks

Explain the laws of reflection of light. Define the incident ray, reflected ray, normal, angle of incidence, and angle of reflection. State the relationship between the angle of incidence and the angle of reflection.

- Correct definition of the incident ray — 1 mark
- Correct definition of the reflected ray — 1 mark
- Correct definition of the normal (perpendicular to the mirror surface at the point of incidence) — 1 mark
- Correct definition of the angle of incidence — 1 mark
- Correct definition of the angle of reflection — 1 mark
- Correct statement of the law of reflection (angle of incidence equals angle of reflection) — 2 marks
- Clear, coherent explanation overall, not just a list of isolated definitions — 1 mark

Part (b) — 7 marks

Draw a complete ray diagram showing the reflection of a light ray from a plane mirror. The diagram must clearly show and label the plane mirror, incident ray, reflected ray, normal, angle of incidence, and angle of reflection.

- Plane mirror correctly drawn — 1 mark
- Incident ray correctly drawn and labelled — 1 mark
- Reflected ray correctly drawn and labelled — 1 mark
- Normal correctly drawn (perpendicular to the mirror at the point of incidence) and labelled — 1 mark
- Angle of incidence correctly marked — 1 mark
- Angle of reflection correctly marked — 1 mark
- Angles are drawn visibly equal and the diagram is accurate overall — 1 mark

Q4: Attempt Both Parts of the following Question [8+7]

Part (a) — 8 marks

What is an algorithm? Explain any three characteristics of a good algorithm. Also state one advantage of writing an algorithm before solving a programming problem.

- Correct definition of an algorithm (a finite, step-by-step procedure or set of instructions to solve a problem) — 2 marks
- Three valid characteristics of a good algorithm correctly explained (e.g. finiteness, unambiguity, well-defined inputs/outputs, effectiveness, language-independence) — 1 mark each, 3 marks total
- One valid advantage of writing an algorithm before programming (e.g. clarifies logic in advance, easier to debug/verify, language-independent, saves development time) — 3 marks

Part (b) — 7 marks

Draw a complete flowchart to find and display the largest of three numbers entered by the user. Use the correct standard flowchart symbols, decision branches, and arrows.

- Correct start and end terminators — 1 mark
- Correct input step for all three numbers — 1 mark
- Correct decision logic (comparisons using diamond/decision symbols) that correctly determines the largest of the three numbers in all cases — 3 marks
- Correct output step displaying the largest number — 1 mark
- Standard flowchart symbols and directional arrows used correctly and consistently throughout — 1 mark

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Total Marks: 35 (Section B: 20, attempt 5 of 7 parts — Section C: 15, attempt 1 of 2 questions)

Note: raw extraction of every listed part in this document sums to more than 35 marks, since Q2 lists all 7 possible parts (28 marks) and both Q3 and Q4 are listed in full (15 marks each) even though only 5 of Q2's parts and one of Q3/Q4 are ever actually attempted or scored. This is expected given the exam's choice structure.